Collaborative Wind Tunnel Flight Investigation

Multi-Variable Functional Scaling of Aerodynamic Stall

$$\alpha_{\text{stall}} = f(\alpha,\, m,\, t/c,\, Re)$$

Platform: NCSSM *WindTunnel lab* (GPU-Accelerated 2D Lattice Boltzmann Simulator)

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Mission Briefing & Research Objective

In aerodynamics, **stall** represents the sudden loss of lift and surge in drag when airflow separates from the upper surface of a wing. Using **Dimensional Analysis**, we can show that the critical stall angle depends on three fundamental dimensionless parameters: airfoil thickness ratio (t/c), camber (m), and the Reynolds number (Re):

$$\alpha_{\text{stall}} = \Phi\left(\frac{t}{c},\, m,\, Re\right), \quad \text{and the peak lift capability} \quad C_{L,\max} = \Psi\left(\frac{t}{c},\, m,\, Re\right). \tag{1}$$

As an aerospace research consortium, our class will systematically explore this multi-variable relationship using the GPU-accelerated **WindTunnel lab**. Three specialized student research teams—each dedicated to one governing dimensionless Π-group—will systematically interrogate their parameter, gather high-precision digital telemetry, and collaborate to construct a unified empirical power-law model for aircraft design.

1 Dimensional Analysis & The Aerodynamics of Stall

1.1 Physical Nature of Aerodynamic Stall

When an airfoil moves through air at an angle of attack α, fluid streams smoothly over both surfaces, generating lift. However, as α increases, air flowing over the upper surface must turn sharply around the nose and decelerate into an increasing pressure hill (known as an *adverse pressure gradient*). At a critical angle called the **stall angle** (α_{stall}), the sluggish air near the surface can no longer overcome this hill. The flow detaches (separates) from the upper surface, spilling into an unsteady, turbulent wake. Lift collapses, drag surges, and the wing stalls.

1.2 Formulating the Physical Problem with Dimensional Analysis

Before running expensive wind tunnel tests or computer simulations, aerospace engineers use **Dimensional Analysis** (the Buckingham Π Theorem) to determine which variables truly govern the physics, reducing complex systems of units into universal, dimensionless parameters.

To determine what governs the stall angle α_{stall}, we list every physical variable that could influence the onset of flow separation:

Physical Variable	**Symbol**	**SI Units**	**Dimensions** $[M, L, T]$
Stall Angle of Attack	α_{stall}	rad or deg	[1] (dimensionless)
Airfoil Chord Length	c	m	$[L]$
Maximum Thickness	t	m	$[L]$
Maximum Camber Height	h	m	$[L]$
Freestream Air Speed	U_∞	m/s	$[LT^{-1}]$
Air Density	ρ	kg/m^3	$[ML^{-3}]$
Air Dynamic Viscosity	μ	Pa · s	$[ML^{-1}T^{-1}]$

1.3 Applying the Buckingham Π Theorem

We have $n = 7$ physical variables expressed in $k = 3$ fundamental dimensions: mass $[M]$, length $[L]$, and time $[T]$. According to the Buckingham Π Theorem, the physical relationship can be expressed by exactly $n - k = 7 - 3 = 4$ independent dimensionless groups (Π-groups).

Selecting the fluid dynamic reference scale (ρ, U_∞, c) as repeating variables, we construct the dimensionless Π-groups:

1. **Target Dimensionless Output ($\Pi_\alpha = \alpha_{\textbf{stall}}$):** The critical stall angle of attack (dimensionless).
2. **Group 1 ($\Pi_1 = t/c$):** The **Airfoil Thickness Ratio** (maximum thickness relative to chord).
3. **Group 2 ($\Pi_2 = h/c = m$):** The **Airfoil Camber Ratio** (maximum camber relative to chord).
4. **Group 3 ($\Pi_3 = \frac{\rho U_\infty c}{\mu} = Re$):** The **Reynolds Number**, which scales fluid inertia (momentum) relative to viscous friction.

Therefore, dimensional analysis proves that the stall angle does not depend on arbitrary combinations of meters, kilograms, or seconds independently. Rather, it is strictly governed by three independent dimensionless groups:

$$\alpha_{\text{stall}} = \mathbf{\Phi}\left(\mathbf{\Pi_1}, \mathbf{\Pi_2}, \mathbf{\Pi_3}\right) = \mathbf{\Phi}\left(\frac{\mathbf{t}}{\mathbf{c}}, \mathbf{m}, \mathbf{Re}\right) \tag{2}$$

These three independent Π-groups directly define the division of labor across our **three specialized student research teams**. Furthermore, during an angle of attack sweep, the dimensionless lift coefficient $C_L = \frac{L}{\frac{1}{2}\rho U_\infty^2 c}$ traces an aerodynamic curve:

$$C_L = g\left(\alpha;\, m,\, \frac{t}{c},\, Re\right) \tag{3}$$

whose peak defines $C_{L,\max}$ occurring at $\alpha = \alpha_{\text{stall}}$.

1.4 Physical Intuition Behind the Dimensionless Groups

- **Angle of Attack (α):** The operational variable. In the attached regime, lift increases linearly ($C_L \approx 2\pi(\alpha - \alpha_0)$). As α increases, the suction peak near the leading edge intensifies until near-wall air runs out of momentum, triggering separation.
- **Airfoil Thickness (t/c):** Controls the roundness of the nose (leading-edge radius $r_{\text{le}} \propto (t/c)^2$). A sharp, thin airfoil ($t/c = 6\%$) forces air around an abrupt corner, causing early, violent bubble separation. A blunt, thick airfoil ($t/c = 18\%$) smooths out this curve, delaying separation to higher angles.

- **Airfoil Camber (m):** Asymmetry shifts the zero-lift angle to negative values ($\alpha_0 \approx -1.1m$), boosting lift at zero degrees. However, extra upper-surface curvature makes the pressure hill steeper, altering when separation occurs.
- **Reynolds Number (Re):** Measures how fast and energetic the fluid is compared to its internal stickiness. High-Re air has high momentum and thinner boundary layers ($\delta \propto 1/\sqrt{Re}$), enabling it to resist separation and climb steeper pressure hills, which delays stall to higher angles.

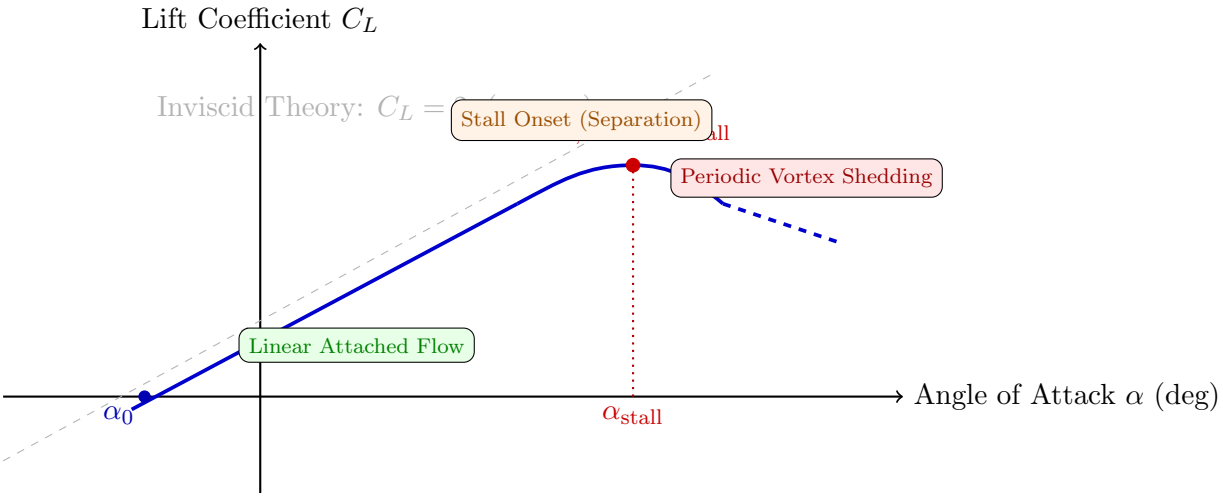


Figure 1: Anatomy of an aerodynamic lift polar: Linear attached regime, non-linear separation onset, peak lift $C_{L,\max}$ at critical stall angle α_{stall}, and post-stall periodic vortex shedding.

2 Experimental Testing Platform: NCSSM *WindTunnel lab*

Tests are conducted using NCSSM's **WindTunnel lab**, a real-time GPU-accelerated numerical wind tunnel powered by the Lattice Boltzmann Method (LBM D2Q9) and Smagorinsky subgrid-scale turbulence modeling.

2.1 Standardized Data Acquisition (DAQ) Protocol

To ensure reproducibility across all student research teams, each data point must be collected according to the **5-Second Integer Vortex Shedding Integration Protocol**:

1. **Geometry Configuration:** Enter the 4-digit NACA profile (e.g., `2412`) in the control panel.
2. **Flow Conditioning:** Set the target Reynolds number (Re) slider and verify that *Smagorinsky LES* is checked.
3. **Angle Dialing:** Adjust α using the slider for coarse sweeping (2.0° increments) and the stepper buttons (`[<]` / `[>]`) for high-resolution refinement (0.5° increments) in the vicinity of stall.
4. **Flow Settling Verification:** Observe the telemetry status indicator. When flow is altered, the indicator displays **Flow Settling**. Wait 3–5 seconds until the badge turns **Flow Settled · Ready**.

5. **Execute 5s Cycle Acquisition:** Click `[Acquire (5s)]` or press hotkey `L`. The system integrates instantaneous momentum-exchange forces across full shedding periods ($\int \sin(\omega t)dt \equiv 0$), completely filtering phase bias.

6. **Telemetry Logging:** The polar graph displays the cycle-averaged C_L, accompanied by dynamic error whiskers representing the shedding envelope $[C_{L,\min}, C_{L,\max}]$.

7. **CSV Export:** Upon completing an angle sweep, click `[CSV]` to save the raw laboratory dataset.

3 Consortium Research Team Assignments (One per Π-Group)

The class is partitioned into **three specialized research teams**, corresponding directly to the three dimensionless Π-groups derived from Buckingham Π analysis:

- **Team 1 ($\Pi_1 = t/c$):** Airfoil Thickness & Nose Curvature Team
- **Team 2 ($\Pi_2 = m$):** Airfoil Camber & Zero-Lift Asymmetry Team
- **Team 3 ($\Pi_3 = Re$):** Reynolds Number & Viscous Scale Team

Each team executes systematic angle of attack sweeps across their assigned Π-group parameter matrix to extract their individual power-law scaling exponent, and additionally acquires one coupled benchmark validation profile to test the consortium's unified general model.

3.1 Team 1: Airfoil Thickness Ratio ($\Pi_1 = t/c$) Team

Primary Objective: Quantify how leading-edge radius and thickness ratio govern stall onset angle, extract the power-law exponent a in $\alpha_{\text{stall}} \propto (t/c)^a$, and evaluate coupled thin-camber validation profile NACA 4406.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Camber $m = 0\%$ (Symmetric Series).
- **Primary Airfoil Sweep Matrix:**
  1. **NACA 0006:** Ultra-thin ($t/c = 6\%$) — Sharp leading edge, high susceptibility to separation bubble burst.
  2. **NACA 0009:** Moderately thin ($t/c = 9\%$) — Classic high-speed profile.
  3. **NACA 0012:** Standard benchmark ($t/c = 12\%$) — NASA calibration standard.
  4. **NACA 0018:** Thick subsonic ($t/c = 18\%$) — High structural depth, blunt nose.
  5. **NACA 0024:** Very thick ($t/c = 24\%$) — Low-speed strut/root section.
- **Angle Sweep Range:** $\alpha \in [-2.0°, +24.0°]$. Sweep in 2.0° steps from $-2°$ to 8°, then 0.5° steps from 8° to 20°.
- **Deliverable Plot 1.1:** Composite polar plot (C_L vs α) overlaying all five thickness ratios.
- **Deliverable Plot 1.2 (Power-Law Evaluation):** Construct a log-log plot of $\ln(\alpha_{\text{stall}})$ vs $\ln(t/c)$ for the attached-flow regime ($t/c \in [0.06, 0.18]$). Determine the thickness power-law exponent a from the slope:

$$\alpha_{\text{stall}} \propto \left(\frac{t}{c}\right)^a. \tag{4}$$

  Identify the peak performance ratio and document why $t/c = 0.24$ departs from this power-law trend.
- **Coupled Validation Benchmark:** Sweep **NACA 4406** ($m = 0.04, t/c = 0.06, Re = 1000$) to provide an independent blind-test case evaluating how high camber acts on an ultra-thin leading edge.

3.2 Team 2: Airfoil Camber Ratio ($\Pi_2 = m$) Team

Primary Objective: Quantify how mean camber line curvature shifts zero-lift angle α_0 and stall angle α_{stall}, extract the power-law exponent b in $\alpha_{\text{stall}} \propto (1+m)^b$, and evaluate coupled thick-camber validation profile NACA 4418.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Thickness $t/c = 12\%$, Maximum Camber Location $p = 0.4c$ (`NACA X412` Family).
- **Primary Airfoil Sweep Matrix:**
  1. **NACA 0012:** Symmetric reference ($m = 0\%$).
  2. **NACA 1412:** Mild camber ($m = 1\%$).
  3. **NACA 2412:** Benchmark camber ($m = 2\%$, typical general aviation wing).
  4. **NACA 4412:** High-lift camber ($m = 4\%$).
  5. **NACA 6412:** Extreme aggressive camber ($m = 6\%$, STOL/cargo section).
- **Angle Sweep Range:** $\alpha \in [-8.0°, +22.0°]$. Sweep in 2.0° steps from $-8°$ to 4°, then 0.5° steps through stall.
- **Deliverable Plot 2.1:** Experimental lift slope verification: verify linear slope $dC_L/d\alpha$ against 2π and plot experimental α_0 vs theoretical prediction $\alpha_0 \approx -1.1m$.
- **Deliverable Plot 2.2 (Power-Law Evaluation):** Construct a plot of normalized stall angle relative to the symmetric baseline, $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs $\ln(1+m)$, where m is fractional camber ($m \in [0.00, 0.06]$). Extract the camber power-law exponent b from the slope:

  $$\alpha_{\text{stall}} \propto (1+m)^b. \tag{5}$$

  Contrast this stall angle sensitivity against the significant gain observed in $C_{L,\max}$ vs m.
- **Coupled Validation Benchmark:** Sweep **NACA 4418** ($m = 0.04, t/c = 0.18, Re = 1000$) to provide an independent blind-test case evaluating how a generous nose radius mitigates aggressive camber stall.

3.3 Team 3: Reynolds Number ($\Pi_3 = Re$) Team

Primary Objective: Quantify how Reynolds number energizes the boundary layer, extract the power-law exponent c in $\alpha_{\text{stall}} \propto Re^c$, and characterize dynamic vortex shedding and inverted stall asymmetry.

- **Fixed Controls:** Standard NACA 2412 airfoil profile ($m = 2\%$, $p = 0.4c$, $t/c = 12\%$).
- **Primary Reynolds Number Sweep Matrix:**
  1. $Re = 300$**:** Viscous-dominated flow (thick laminar boundary layer, micro-aerial scale).
  2. $Re = 600$**:** Low Reynolds number transitional regime.
  3. $Re = 1200$**:** Standard baseline tunnel scale.
  4. $Re = 2500$**:** Moderate Reynolds number with elevated momentum transport.
  5. $Re = 5000$**:** High Reynolds number testing (energized boundary layer).
- **Angle Sweep Range:** $\alpha \in [-4.0°, +22.0°]$ for each Reynolds number. Focus 0.5° resolution in the $[8°, 18°]$ stall zone.

- **Deliverable Plot 3.1:** Family of $C_L(\alpha)$ curves showing the progression of stall softening and delay with increasing Re.
- **Deliverable Plot 3.2 (Power-Law Evaluation):** Construct a log-log scaling plot of $\ln(\alpha_{\text{stall}})$ vs $\ln(Re)$ across all five test points. Determine the Reynolds power-law exponent c from the slope:

$$\alpha_{\text{stall}} \propto Re^{c}. \tag{6}$$

  Relate exponent c to the momentum diffusion of boundary layer thinning ($\delta \propto Re^{-1/2}$).
- **Dynamic Shedding & Inverted Benchmark:** Execute a continuous bidirectional sweep on **NACA 2412** from $\alpha = -16.0°$ to $+20.0°$. Locate positive stall $(\alpha^{+}_{\text{stall}}, C^{+}_{L,\max})$ and inverted negative stall $(\alpha^{-}_{\text{stall}}, C^{-}_{L,\min})$, and plot shedding amplitude $\Delta C_L = C_{L,\max} - C_{L,\min}$ (error whiskers) past stall.

4 Classroom Consortium: Multi-Variable Data Pooling

Following data acquisition, each team will deposit their calibrated data points into the **Consortium Master Telemetry Database**. Every student will then have access to the complete parametric matrix for power-law synthesis and model validation.

Team	Airfoil / Case	m	t/c	Re	Meas. α_{stall} (°)	Meas. $C_{L,\max}$
Team 1: Thickness ($\Pi_1 = t/c$) Matrix ($m = 0$, $Re = 1000$)						
1	NACA 0006	0.00	0.06	1000	_____	_____
1	NACA 0009	0.00	0.09	1000	_____	_____
1	NACA 0012 (Symmetric Ref.)	0.00	0.12	1000	_____	_____
1	NACA 0018	0.00	0.18	1000	_____	_____
1	NACA 0024	0.00	0.24	1000	_____	_____
Team 2: Camber ($\Pi_2 = m$) Matrix ($t/c = 0.12$, $Re = 1000$)						
2	NACA 1412	0.01	0.12	1000	_____	_____
2	NACA 2412 (GA Benchmark)	0.02	0.12	1000	_____	_____
2	NACA 4412	0.04	0.12	1000	_____	_____
2	NACA 6412	0.06	0.12	1000	_____	_____
Team 3: Reynolds Number ($\Pi_3 = Re$) Matrix (NACA 2412)						
3	NACA 2412 ($Re = 300$)	0.02	0.12	300	_____	_____
3	NACA 2412 ($Re = 600$)	0.02	0.12	600	_____	_____
3	NACA 2412 ($Re = 1200$)	0.02	0.12	1200	_____	_____
3	NACA 2412 ($Re = 2500$)	0.02	0.12	2500	_____	_____
3	NACA 2412 ($Re = 5000$)	0.02	0.12	5000	_____	_____
Consortium Benchmark & Coupled Validation Targets						
1	NACA 4406 (Team 1 Coupled Validation)	0.04	0.06	1000	_____	_____
2	NACA 4418 (Team 2 Coupled Validation)	0.04	0.18	1000	_____	_____
3	NACA 2412 (Team 3 Inverted Benchmark)	0.02	0.12	1000	_____	_____

Table 1: Consortium Master Parametric Telemetry Table organized by Π-group research teams.

5 Analysis & Discussion Prompts

Each research team will author a technical report evaluating their isolated Π-group power-law dependency and synthesizing the global consortium findings. Address the following analytical prompts:

1. **Team 1 Analysis: Thickness Power-Law ($\Pi_1 = t/c$) & Aerodynamic Limits:** Evaluate Team 1's isolated parameter sweep.
   - Present the log-log plot of $\ln(\alpha_{\text{stall}})$ vs $\ln(t/c)$ for $t/c \in [0.06, 0.18]$.
   - Determine the empirical thickness power-law exponent a in $\alpha_{\text{stall}} \propto (t/c)^a$.
   - Physically explain the stall mechanism transition between NACA 0006 (thin-airfoil bubble bursting) and NACA 0018 (progressive trailing-edge stall).
   - Why does increasing thickness beyond $t/c \approx 0.18$ cause a departure from this power law?

2. **Team 2 Analysis: Camber Power-Law ($\Pi_2 = m$) & Lift Capability:** Evaluate Team 2's isolated parameter sweep.
   - Present the plot of normalized stall angle $\ln(\alpha_{\text{stall}}/\alpha_{\text{stall},0})$ vs $\ln(1+m)$.
   - Determine the empirical camber power-law exponent b in $\alpha_{\text{stall}} \propto (1+m)^b$.
   - Contrast the effect of camber on $C_{L,\max}$ (significant boost) versus its effect on α_{stall} (mild shift). Why does NACA 6412 achieve higher $C_{L,\max}$ than NACA 0012 despite an intensified suction peak?
   - Verify Thin Airfoil Theory: calculate the lift curve slope $dC_L/d\alpha$ in the linear regime and compare to 2π. Calculate the zero-lift angle α_0 and compare to theoretical prediction $\alpha_0 \approx -1.1m$.
3. **Team 3 Analysis: Reynolds Number Power-Law ($\Pi_3 = Re$) & Viscous Momentum:** Evaluate Team 3's isolated parameter sweep.
   - Present the log-log scaling plot of $\ln(\alpha_{\text{stall}})$ vs $\ln(Re)$ across all five test points.
   - Determine the empirical Reynolds power-law exponent c in $\alpha_{\text{stall}} \propto Re^c$.
   - Relate your exponent c to the laminar boundary layer thickness scaling $\delta/c \propto Re^{-1/2}$. Why does an energized, thinner boundary layer delay flow separation to higher angles?
4. **Post-Stall Dynamics, Inverted Asymmetry & Force Vector Behavior:** Observe the real-time vector overlay ($\vec{L}, \vec{D}, \vec{R}$) and smoke streamlines across stall.
   - How does the orientation of the resultant force vector $\vec{R}$ rotate as the airfoil transitions into deep stall?
   - In deep stall, how does the shedding oscillation envelope $[C_{L,\min}, C_{L,\max}]$ relate to the formation and shedding of von Kármán vortex street structures in the wake?
   - Using Team 3's bidirectional sweep, contrast positive stall $(\alpha^+_{\text{stall}}, C^+_{L,\max})$ against inverted stall $(\alpha^-_{\text{stall}}, C^-_{L,\min})$ on NACA 2412.
5. **Consortium Synthesis: Proposing the Unified General Stall Law:** Combine the individual power-law dependencies determined separately by Teams 1, 2, and 3 to propose a general multiplicative formula for the stall angle:

$$\alpha_{\text{stall}}\left(\mathbf{m},\, \frac{\mathbf{t}}{\mathbf{c}},\, \mathbf{Re}\right) = \mathbf{K} \cdot \left(\frac{\mathbf{t}}{\mathbf{c}}\right)^{\mathbf{a}} \cdot (\mathbf{1}+\mathbf{m})^{\mathbf{b}} \cdot \mathbf{Re}^{\mathbf{c}} \tag{7}$$

   - **Exponent Assembly:** Substitute the numerical values of exponents a, b, and c obtained by each Π-group team.
   - **Calibration Constant:** Using the baseline benchmark airfoil (e.g., NACA 0012 or NACA 2412 at $Re = 1000$), calculate the leading constant K:

$$K = \frac{\alpha_{\text{stall, base}}}{\left(\frac{t}{c}\right)^a_{\text{base}} \cdot (1+m_{\text{base}})^b \cdot Re^c_{\text{base}}}. \tag{8}$$

   - **Model Validation against Coupled Benchmarks:** Use your general formula to predict the stall angle for the coupled validation test cases:
     (a) **NACA 4406:** ($m = 0.04$, $t/c = 0.06$, $Re = 1000$)
     (b) **NACA 4418:** ($m = 0.04$, $t/c = 0.18$, $Re = 1000$)

     Calculate the percentage error between your model's prediction and the experimentally measured stall angle:

$$\text{Percent Error} = \left|\frac{\alpha_{\text{pred}} - \alpha_{\text{meas}}}{\alpha_{\text{meas}}}\right| \times 100\%. \tag{9}$$

- **Physical Limitations:** Discuss how well the combined product form predicts coupled geometry cases. Under what aerodynamic conditions does the assumption of independent power-law scaling begin to break down?

6 Grading Rubric (100 Points Total)

Criterion	Performance Expectations	Points
Experimental Protocol & DAQ Rigor	Correct execution of 5s burst integration protocol; flow settled state verified; appropriate angle density (0.5° near stall); complete CSV exports.	20
Data Visualization & Polar Curves	Clear, publication-grade polar plots (C_L vs α, C_D vs α, L/D vs α) with error whiskers, stall points clearly identified, theoretical references included.	20
Physical Mechanisms & Fluid Dynamics	Sound, thorough physical reasoning explaining separation, suction peaks, boundary layer energy, vortex shedding, and thin vs trailing-edge stall.	25
Consortium Synthesis & Power-Law Model	Accurate exponent extraction from log-log power-law plots across all three Π-groups; successful synthesis of general formula $\alpha_{\text{stall}} = K(t/c)^a(1+m)^b Re^c$; validation and error analysis against coupled test cases.	20
Technical Communication & Formatting	Clear engineering report organization; proper unit annotations; professional tables, diagrams, and executive summary.	15
Total		**100**

Table 2: Laboratory Evaluation and Assessment Rubric.